

density (27.6.4) by an amount proportional to $\text{Im} \sum_{\alpha\beta} W_{A\alpha L} W_{\beta L}$, which as we saw in Section 27.3 is a spacetime derivative, and therefore can have no effect in perturbation theory. This translation symmetry prevents X from appearing anywhere in the effective Lagrangian density (27.6.5) except where it appears in the original Lagrangian density (27.6.4). We therefore conclude that f_λ is independent of X , while $h_{\lambda AB}$ consists of a Φ -independent term proportional to $X\delta_{AB}$, plus a term that is independent of X . That is,

$$\mathcal{B}_\lambda(\Phi, W_L, X, Y) = Y f_\lambda(\Phi) + \sum_{\alpha\beta AB} \epsilon_{\alpha\beta} W_{A\alpha L} W_{\beta L} \left[c_\lambda \delta_{AB} X + \ell_{\lambda AB}(\Phi) \right], \quad (27.6.7)$$

where c_λ is a real cut-off-dependent constant.

The point of introducing the external auxiliary superfields X and Y is that, by giving them suitable values, we can make use of weak-coupling approximations to determine the coefficients in Eq. (27.6.7). If we set the spinor and auxiliary components of X and Y equal to zero, and take their scalar components x and y to approach infinity and zero, respectively, then the gauge coupling constant vanishes as $1/\sqrt{x}$, and all Yukawa and scalar couplings derived from the superpotential vanish as y . In this limit, the only graph that contributes to the term in (27.6.7) proportional to Y has a single vertex arising from the term $2 \text{Re} [Y f(\Phi)]_{\mathcal{F}}$ in Eq. (27.6.4), so

$$f_\lambda(\Phi) = f(\Phi). \quad (27.6.8)$$

Also, with $Y = 0$ there is a conservation law which requires every term in $\mathcal{L}_\lambda^\#$ to have equal numbers of Φ s and $\Phi^\dagger$ s, so since $\Phi^\dagger$ cannot occur in $\ell_{\lambda AB}$, neither can Φ . Gauge invariance then requires the constant $\ell_{\lambda AB}$ to be proportional to δ_{AB} for a simple group:

$$\ell_{\lambda AB} = \delta_{AB} L_\lambda. \quad (27.6.9)$$

Now, since gauge propagators go as $1/x$ while pure gauge interactions go as x and scalar propagators and interactions are x -independent, with $y = 0$ the number of powers of x in a diagram with V_W pure gauge boson vertices, I_W internal gauge boson lines, and any number of scalar-gauge boson vertices and scalar propagators is

$$N_x = V_W - I_W. \quad (27.6.10)$$

The number of loops is given by

$$L = I_W + I_\Phi - V_W - V_\Phi + 1, \quad (27.6.11)$$

where I_Φ is the number of internal Φ lines and V_Φ is the number of Φ - V interaction vertices. All the Φ - V vertices have two Φ lines attached, so with no external Φ lines I_Φ and V_Φ are equal, and therefore cancel in